



CIRCL
Computer Incident
Response Center
Luxembourg



GCVE.eu

Vulnerability Lookup and GCVE

A Decentralized Approach to Vulnerability Publishing and Management

<https://vulnerability-lookup.org> – <https://gcve.eu/>

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CIRCL <https://www.circl.lu>

Vulnerability-Lookup

Who is behind Vulnerability-Lookup?



Vulnerability-Lookup¹ is an **Open Source** project led by **CIRCL**.
It is co-funded by **CIRCL** and the **European Union**².
Used by many organisations including CSIRTs and ENISA (EUVD).
db.GCVE.eu is also a vulnerability-lookup instance.



vulnerability
-lookup

¹<https://www.vulnerability-lookup.org>

²<https://github.com/ngsoti>

What is Vulnerability-Lookup?

- A unified place to **search**, **triage**, and **track** software and product vulnerabilities from different sources.
- Brings together vulnerability information, correlation of identifiers (e.g., CVE, GHSA, OSV), references, timelines, and risk signals in one view.
- Designed for **CSIRTs**, **SOCs**, **vulnerability managers**, and **developers** in mind.
- **Web UI first** with strong **API** and automation options.
- Support a complete **CVD process management**³ along with the ability to fork vulnerability information⁴. Distributed GNA directory and GCVE automatic publication and pulling.

³CNA Program, GCVE GNA Publication

⁴Implemented before the GNA initiative!

Recent vulnerabilities

CVE List v5

NVD

FKIE NVD

GitHub

PySec

emb3d

GSD

OpenSSF Malicious Packages

CSAF CERT-Bund

CSAF NCSC-NL

CSAF Trustsource

CSAF Siemens

CSAF Red Hat

CSAF CISA

CSAF CISCO

CSAF Sick

CSAF Nozomi Networks

CSAF Open-Xchange

CSAF Microsoft

CSAF ABB

VARIoT

JVNDB

Tailscale

CSAF Suse

CSAF OpenSuse

CSAF Schneider Electric

CNVD

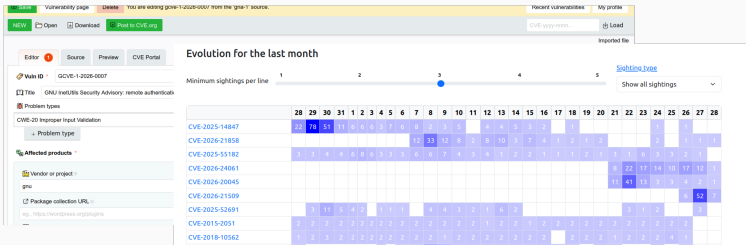
CERT FR Avis

CERT FR Alerte

CIRCL

AHA!

ID	CVSS	Description	Vendor	Product	Published	Updated
CVE-2025-36911	7.1 (3.1)	In key-based pairing, there is a possible ID due ...	Google	Android	12 days ago	39 minutes ago
CVE-2026-24858	9.4 (3.1)	An Authentication Bypass Using an Alternate Path ...	Fortinet	FortiAnalyzer	10 hours ago	39 minutes ago



⁵<https://db.gcve.eu> - <https://vulnerability.circl.lu/>

Origin of Vulnerability-Lookup

- `cve-search`⁶ is an open-source tool initially developed in late 2012, focusing on maintaining a **local** CVE database.
- `cve-search` is widely used as an **internal** tool.
- The design and scalability of `cve-search` are limited. Our operational public instance has reached a hard limit of 20,000 queries per second.
- Vulnerability sources have **diversified**, and the **NVD CVE is no longer the sole source** of vulnerability information.

⁶<https://github.com/cve-search/cve-search>

Initial (Solved) Challenges

- **Volume of data:** Handling a substantial dataset and heavy network traffic, currently over **2,081,173** security advisories and more than **190,955** sightings collected (from MISP, Nuclei, MetaSploit, Social Networks) in approximately a year⁷.
- **Flexibility:** Balancing ongoing development with legacy issues while designing a future-proof architecture. It's complex and yes, sometimes chaotic⁸.
- **Robustness:** Validating data even when external entities don't comply with their own JSON schemas. It's not always pretty.
- **Fast lookup:** Rapidly correlating identifiers across **diverse sources**, including unpublished advisories.

⁷The first sighting on Exploit-DB dates back 26 years.

⁸We enjoy challenges, especially when they lead to practical solutions.

- **CVD process:** Developing an open-source tool to support Coordinated Vulnerability Disclosure.⁹
- **Vulnerability numbering:** Enabling a distributed approach via the Global CVE Allocation System.¹⁰
- **Scoring vulnerabilities:** Aggregating large volumes of observations from diverse advisory types to improve scoring. Automatic **weighting** of sightings by source or type (e.g., exploited, PoC available) and **detection of type** independent of the source.
- **Data quality:** Transforming messy data into clean datasets suitable for model training and analysis¹¹.

⁹Aligned with NIS 2 and the Cyber Resilience Act.

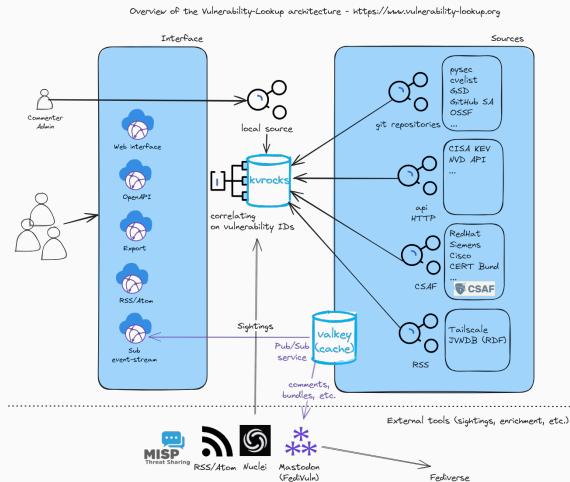
¹⁰<https://gcve.eu>

¹¹<https://github.com/vulnerability-lookup/VulnTrain>

- **CVE Program** Git
- **NIST NVD CVE** API 2.0
- **Fraunhofer FKIE CVE** Git
- **GCVE** API
GNA already publishing
- **GitHub Advisory Database** Git
- **Python Advisory Database** Git
- **Cloud Security Alliance - GSD** Git
- **VARIoT** API
- **CSAF 2.0 (HTTP CSAF)**
CERT-Bund, Cisco, Siemens, Red Hat, Suse,
OpenSuse, Microsoft, NCSC-NL, CISA, ABB, SE, etc.
- **Japan - JVN DB** HTTP
- **China - CNVD** HTTP
- **CERT-FR** HTTP
- **Tailscale** RSS
- **CISA KEV** HTTP
- **EU KEV** HTTP
- **Growing...**

Vulnerability-Lookup High-Level Architecture

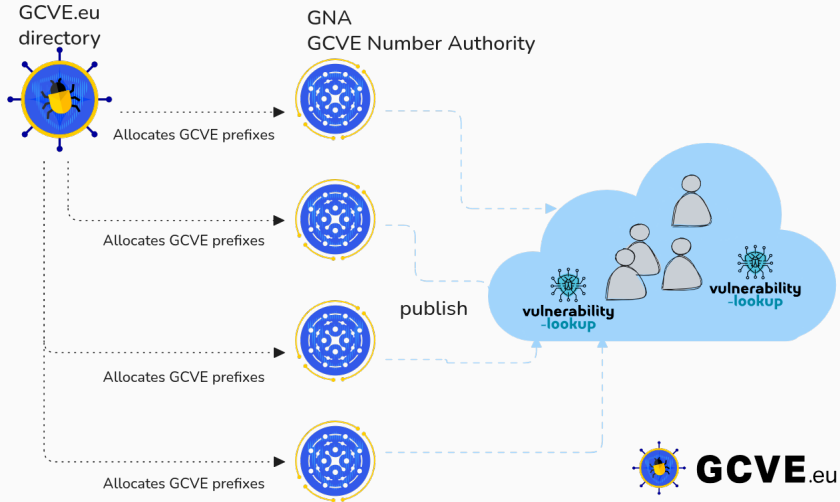
- Collects and aggregates vulnerability data.
- Preserves data integrity (**never alters the original content**).
- Harmonizes data using kvrocks.
- **Correlates** information across multiple sources (CVE, PySec, etc.).
- Provides both APIs and a Web interface.
- Supports advanced queries and filtering.



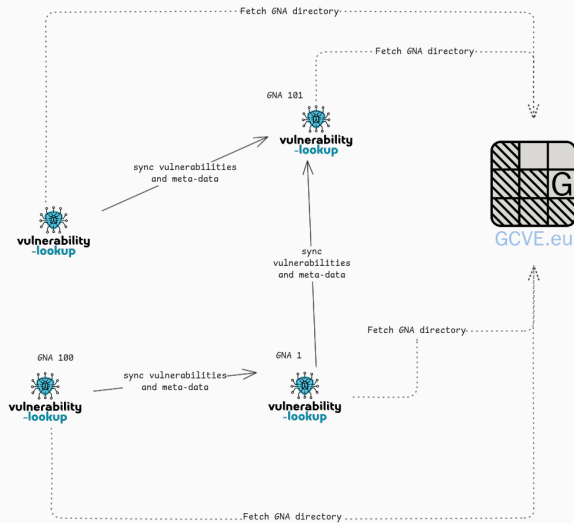
Global CVE Allocation System

- The primary role of GCVE is to provide **globally unique identifiers** to GCVE Numbering Authorities (GNAs).
- **GNAs operate autonomously**, with full control over how they assign and manage identifiers.
- **GCVE publishes Best Current Practices (BCPs)** on directory management, Coordinated Vulnerability Disclosure (CVD), and publication protocols.
- GCVE maintains and publishes the **official directory of all GNAs**, including their publication endpoints.

Overview



A Distributed Network



Who Can Be a GNA?

- You are an existing CNA recognized by the CVE Program.
- You are not a CNA, but **meet at least one of the following conditions**¹²:
 - You are a registered CSIRT or CERT listed on FIRST.org, part of the EU CSIRTs Network, or a member of TF-CSIRT.
 - You are a software, hardware, or service provider that regularly discloses vulnerabilities affecting your own products or services, and you have an official CPE vendor name assigned.
 - You have a public vulnerability disclosure policy and maintain a publicly accessible source for newly disclosed vulnerabilities in GCVE-BCP 05 format.

¹²<https://gcve.eu/about/#eligibility-and-process-to-obtain-a-gna-id>

Benefits of Becoming a GNA

- **Fast, straightforward onboarding:** as soon as the eligibility criteria are met, registration is quick and simple.
- **Flexible identifier usage:** you may publish new CVE entries immediately and apply your assigned prefix to both current and historical identifiers.
- **Autonomy over publication:** each GNA determines for itself what constitutes a vulnerability and what information is made public.
- **Incremental adoption of BCPs:** additional GCVE Best Current Practices can be adopted over time; implementing every BCP is encouraged but not mandatory.

Best Current Practices (BCPs)

ID	Title	Status	Ver.	Published
GCVE-BCP-01	Signature Verification of the Directory File	PUB	1.1	25 Apr 2025
GCVE-BCP-02	Practical Guide to Vulnerability Handling and Disclosure	PUB	1.5	25 Jan 2026
GCVE-BCP-03	Decentralized Publication Standard	<i>Draft</i>	1.0	10 Jun 2025
GCVE-BCP-04	Recommendations and Best Practices for ID Allocation	PUB	1.3	02 Oct 2025
GCVE-BCP-05	GCVE Vulnerability Format (Updated CVE Record Format)	<i>Draft</i>	1.6	02 Jan 2026
GCVE-BCP-07	Known Exploited Vulnerability - KEV Assertion Format	<i>Draft</i>	1.5	27 Jan 2026

Draft and public documents are always open for public review.

How to Publish with GCVE in 4 Steps

- 1. Install vulnerability-lookup or generate a feed compliant with GCVE-BCP-05.
- 2. Apply to become a GCVE Numbering Authority (GNA) and submit your feed URL.
- 3. Your feed is now discoverable and consumable by all software in the GCVE network.
- 4. Make some tea.

Closing

Ongoing Development

- Deeper analysis of content and context around sightings and exploited vulnerabilities¹³.
- Insight generation based on patch datasets.
- Cross-source CPE/PURL guessing.
- KEV support with synchronization across all vulnerability-lookup instances.
- Full-text search capabilities across all data sources.
- Synchronization between multiple vulnerability-lookup instances, enabling **real-time sightings across instances**.



The project is evolving rapidly — we actively welcome feedback¹⁴ and feature suggestions!

¹³<https://github.com/vulnerability-lookup/TARDISsight>

¹⁴<https://discourse.ossbase.org/c/gcve/14>

<https://discourse.ossbase.org/c/vulnerability-lookup/6>

References

🏠 <https://www.vulnerability-lookup.org> - <https://gcve.eu/>

📖 <https://vulnerability.circl.lu> - <https://db.gcve.eu/>

🐙 <https://github.com/vulnerability-lookup/vulnerability-lookup>

📧 <https://social.circl.lu/@circl> - <https://social.circl.lu/@gcve>

Thank you for your attention

- Issues, new sources of advisories or ideas:
<https://github.com/vulnerability-lookup/vulnerability-lookup>
- For support and questions, contact: info@circl.lu
- If you want to become a GNA, gna@gcve.eu
- GCVE [@gvce@social.circl.lu](https://twitter.com/gvce)